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TESS Data Release Notes:
Sectors 14 – 86, Multi-sector Search, DR122

Joseph D. Twicken, Douglas A. Caldwell, Benjamin M. Tofflemire, Marziye Jafariyazani
SETI Institute, Mountain View, California

Michael M. Fausnaugh
Texas Tech University, Lubbock, Texas

Jon M. Jenkins
NASA Ames Research Center, Moffett Field, California

Roland Vanderspek
Kavli Institute for Astrophysics and Space Science, Massachusetts Institute of Technology, Cambridge, Massachusetts

Stephanie Striegel
SETI Institute, Mountain View, California

Eric B. Ting
NASA Ames Research Center, Moffett Field, California

Rebekah Hounsell
University of Maryland Baltimore County, Baltimore, Maryland
NASA Goddard Space Flight Center, Greenbelt, Maryland

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Acknowledgements

These Data Release Notes provide information on the processing and export of data from the Transiting Exoplanet Survey Satellite (TESS). This data release is a combined, multi-sector transit search only. The underlying data products from individual observing sectors have been previously released. The data products included in this data release are the Data Validation (DV) reports, time series, and associated xml files for the threshold crossing events (TCEs) found by searching a combined data set including data from multiple observing sectors.

These data products were generated by the TESS Science Processing Operations Center (SPOC, Jenkins et al., 2016) at NASA Ames Research Center from data collected by the TESS instrument, which is managed by the TESS Payload Operations Center (POC) at Massachusetts Institute of Technology (MIT). The format and content of these data products are documented in the Science Data Products Description Document (SDPDD)[1]. The SPOC science algorithms are based heavily on those of the Kepler Mission science pipeline, and are described in the Kepler Data Processing Handbook (Jenkins, 2020)[2]. The Data Validation algorithms are documented in Twicken et al. (2018) and Li et al. (2019). The TESS Instrument Handbook (Vanderspek et al., 2018) contains more information about the TESS instrument design, detector layout, data properties, and mission operations.

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This report is available in electronic form at
https://archive.stsci.edu/tess/

[1]https://archive.stsci.edu/missions/tess/doc/EXP-TESS-ARC-ICD-TM-0014-Rev-F.pdf
[2]https://archive.stsci.edu/kepler/manuals/KSCI-19081-003-KDPH.pdf

1 Data

TESS Data Release DR122 consists of results from a transiting planet search conducted with the combined data from Sectors 14 through 26 (TESS Year 2 – northern ecliptic pointings), Sectors 40 through 41 (TESS Year 4 – northern ecliptic pointings), Sectors 42 through 45 (TESS Year 4 – ecliptic pointings), Sectors 47 through 60 (TESS Years 4 and 5 – northern ecliptic pointings), Sectors 70 through 72 (TESS Year 6 – ecliptic pointings), and Sectors 73 through 86 (TESS Years 6 and 7 – northern ecliptic pointings). Figure 1 shows the Right Ascension (RA) and Declination (Dec) of all two-minute targets, color-coded by the number of sectors for which each target was observed. Targets with new data in any of Sectors 79 through 86 that were observed in at least one other previous sector were subjected to a multi-sector planet search (see Data Releases 23, 30, 34, 40, 61, 63, 68, 69, 75, 81, 87, 104 and 113 for previous multi-sector planet searches involving Sectors 14 – 86 data). For Sectors 14 – 26, the data correspond to the 2-minute reprocessed data from DR73, while for Sectors 40 – 86 the data are the same 2-minute cotrended light curves presented in previous single sector data releases. Table 1 provides basic information and data release numbers for the observations of each sector. The observations span a 1979 day period; Figure 2 shows the times with TESS observations and gaps.

Table 2 summarizes the total number of targets with multi-sector data for this data release. A supplemental table[3] lists the targets searched, including a string indicating which sectors the target was observed in, whether or not the target produced a TCE, and whether or not the target completed DV analysis.

The multi-sector planet search is performed using a binned data mode. The binned data mode combines data from five 2-minute cadences into a single super-cadence with a 10 minute duration. Binning permits reduced pipeline runtime and memory usage, while maintaining sensitivity to planet detection and characterization. The archival DV reports and time series provide figures and data that are binned at the 10 minute super-cadence. Super-cadence boundaries are not the same for all targets; that is, individual targets have super-cadence start/end times that depend on the sector in which they were first observed (the start time of the first super-cadence for a given target is the start time of the first sector in which the target was observed).

[3]https://archive.stsci.edu/missions/tess/catalogs/targetinfo/tess_multisector_14_86_drn122_targetinfo_v01.txt

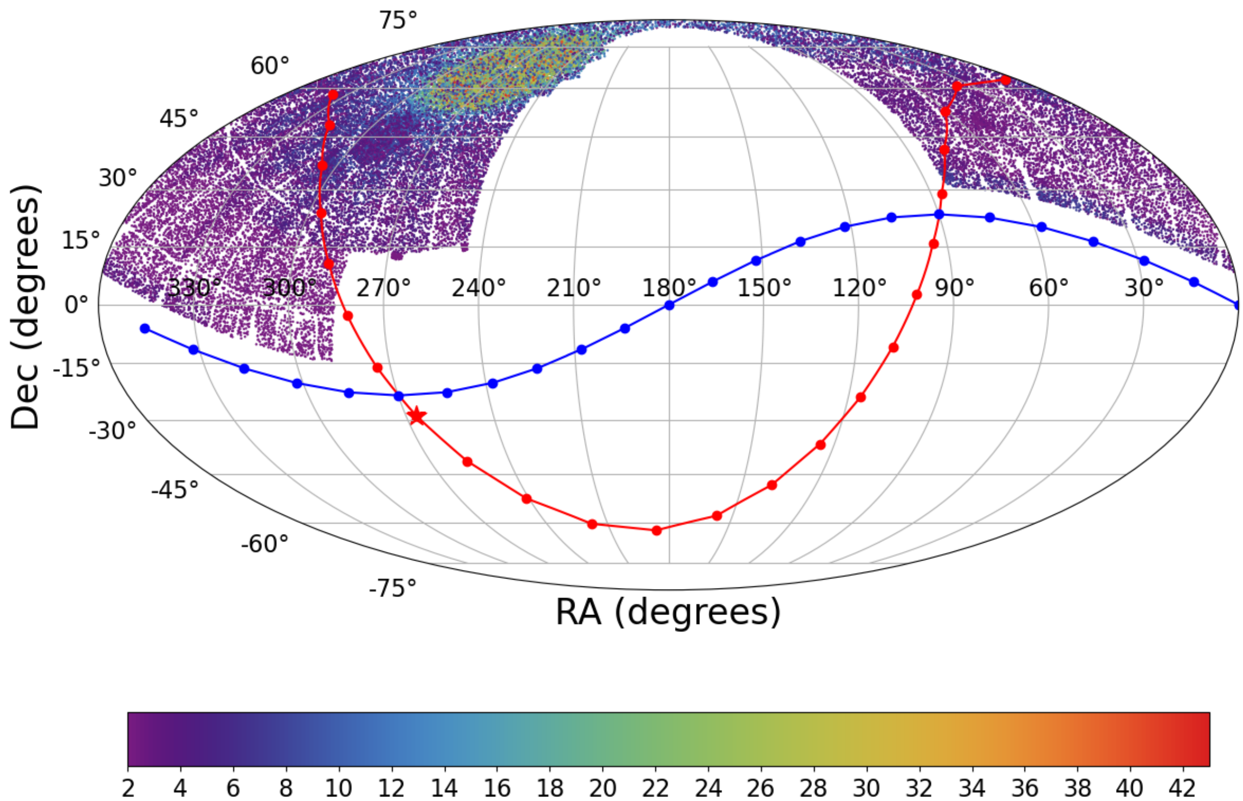

Figure 1: Right Ascension and Declination for TESS two-minute targets included in this multi-sector search, color-coded by the number of sectors in which that target was observed. The galactic and ecliptic planes are indicated by the red and blue lines, respectively.

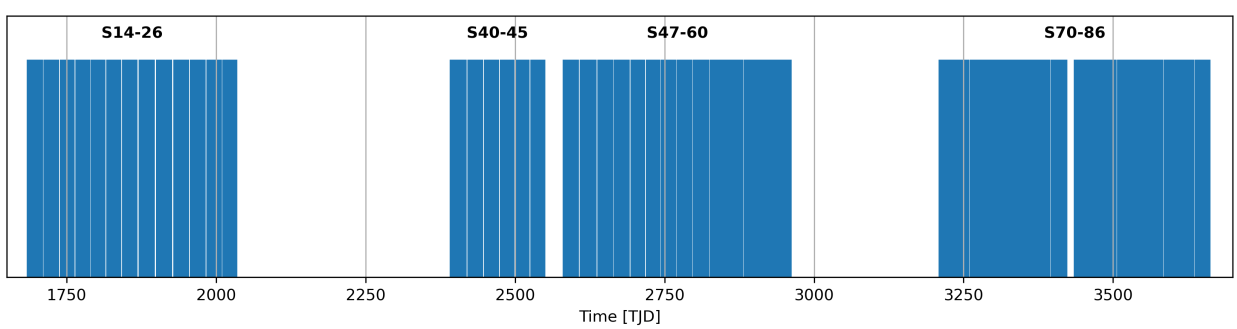

Figure 2: Observation times for sectors included in this multi-sector transit search run. Note: most targets were observed for only a subset of these sectors.

2 Transit Search and Data Validation

The light curves of 40681 targets observed in Sectors 14 through 86 were subjected to the transit search in TPS. Figure 3 shows the 1-hour CDPP for the combined light curves of the 40681 targets. Threshold Crossing Events (TCEs) at the $7.1\,\sigma$ level were generated for 10404 of these targets. A search for additional TCEs in potential multiple planet systems was conducted in DV through calls to TPS. A total of 19521 TCEs were identified in the SPOC pipeline on 10402 unique target stars that completed analysis in DV. There were 2 targets (355257343 and 356108698) which did not complete analysis in DV; the TCEs associated with these targets are not available in the archived data products. Targets which failed to successfully complete DV are flagged in the supplemental target information table. Table 3 provides a breakdown of the number of TCEs by target. Note that targets with large numbers of TCEs are likely to include false positives.

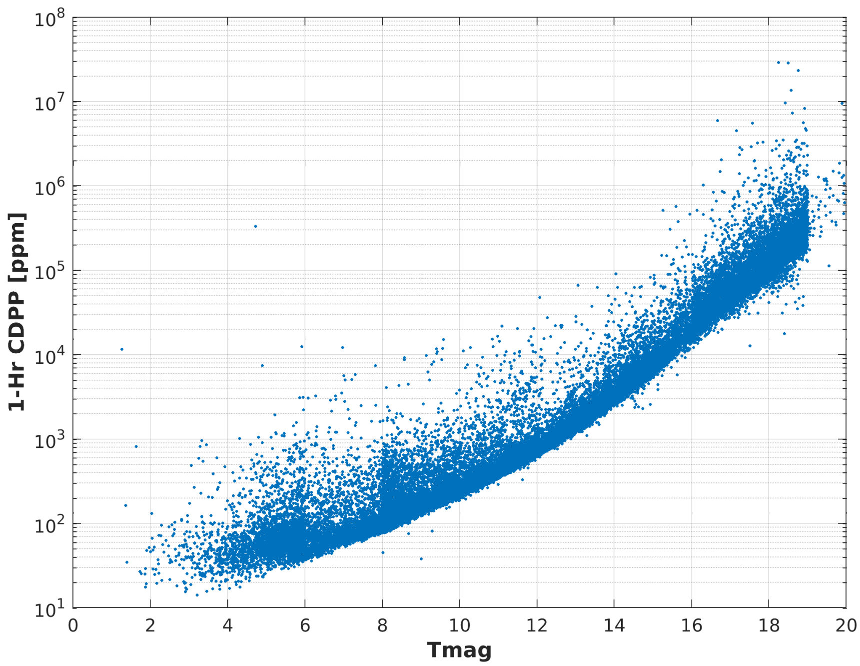

Figure 3: 1-hour CDPP. The points are RMS CDPP measurements for the 40681 light curves from the Sectors 14 – 86 multi-sector search plotted as a function of TESS magnitude.

Figure 4 gives the distribution in period–transit depth space of the TCEs found in the multi-sector search. The top panel shows the distribution of orbital periods for the TCEs. After declining for periods out to 10 days, the distribution shows several preferred orbital periods towards the longest period allowed ($\lesssim$1979 day) while requiring at least two transit events.

The true orbital period of TCEs reported here may be ambiguous due to gaps in observations. TPS requires a minimum of two transit-like events to trigger a TCE, but does not require that all transits implied by the estimated period correspond to observed data. The excess of TCEs at various periods in Figure 4 are attributable partially to the ambiguities introduced by the data gaps. Systematic noise signals contribute to the bulk of these TCE excesses.

The vertical histogram in the right panel of Figure 4 shows the distribution of transit depths derived from limb-darkened transiting planet model fits for TCEs. The model transit depths range down to the order of 10 ppm, but the bulk of the transit depths are considerably larger.

Table 1: Sectors Searched

Sector #	Physical Orbits	Start TJD[a]	End TJD	Data Release #
14	35,36	1683.348	1710.204	73
15	37,38	1711.359	1737.409	73
16	39,40	1738.647	1763.319	73
17	41,42	1764.679	1789.694	73
18	43,44	1790.651	1815.030	73
19	45,46	1816.077	1841.148	73
20	47,48	1842.498	1868.822	73
21	49,50	1870.429	1897.780	73
22	51,52	1899.301	1926.493	73
23	53,54	1928.100	1954.875	73
24	55,56	1955.790	1982.280	73
25	57,58	1983.627	2009.305	73
26	59,60	2010.262	2035.134	73
40	87,88	2390.648	2418.853	57
41	89,90	2419.984	2446.578	59
42	91,92	2447.686	2473.159	60
43	93,94	2474.162	2498.888	62
44	95,96	2500.183	2524.440	64
45	97,98	2525.502	2550.624	65
47	101,102	2579.796	2606.940	67
48	103,104	2607.928	2635.985	70
49	105,106	2637.467	2664.314	71
50	107,108	2665.266	2691.509	72
51	109,110	2692.942	2717.534	74
52	111,112	2718.630	2743.076	76
53	113,114	2743.991	2768.978	77
54	115,116	2769.894	2796.128	78
55	117,118	2797.095	2824.263	79
56	119,120	2825.252	2853.138	80
57	121,122	2853.138	2882.114	82
58	123,124	2882.324	2910.048	83
59	125,126	2910.259	2936.687	84
60	127,128	2936.898	2962.583	85
70	147,148	3208.349	3233.819	99

71	149,150	3234.030	3259.968	101
72	151,152	3260.178	3285.582	102
73	153,154	3285.793	3312.649	103
74	155,156	3312.859	3339.568	105
75	157,158	3339.780	3367.482	106
76	159,160	3367.691	3394.777	107
77	161,162	3395.481	3423.549	108
78	163,164	3434.264	3452.322	110
79	165,166	3452.533	3479.673	111
80	167,168	3479.883	3506.336	112
81	169,170	3506.546	3533.177	114
82	171,172	3533.387	3559.246	115
83	173,174	3559.421	3584.376	116
84	175,176	3584.584	3610.341	117
85	177,178	3610.549	3636.045	118
86	179,180	3636.255	3662.827	119

[a] TJD = TESS JD = JD - 2,457,000.0

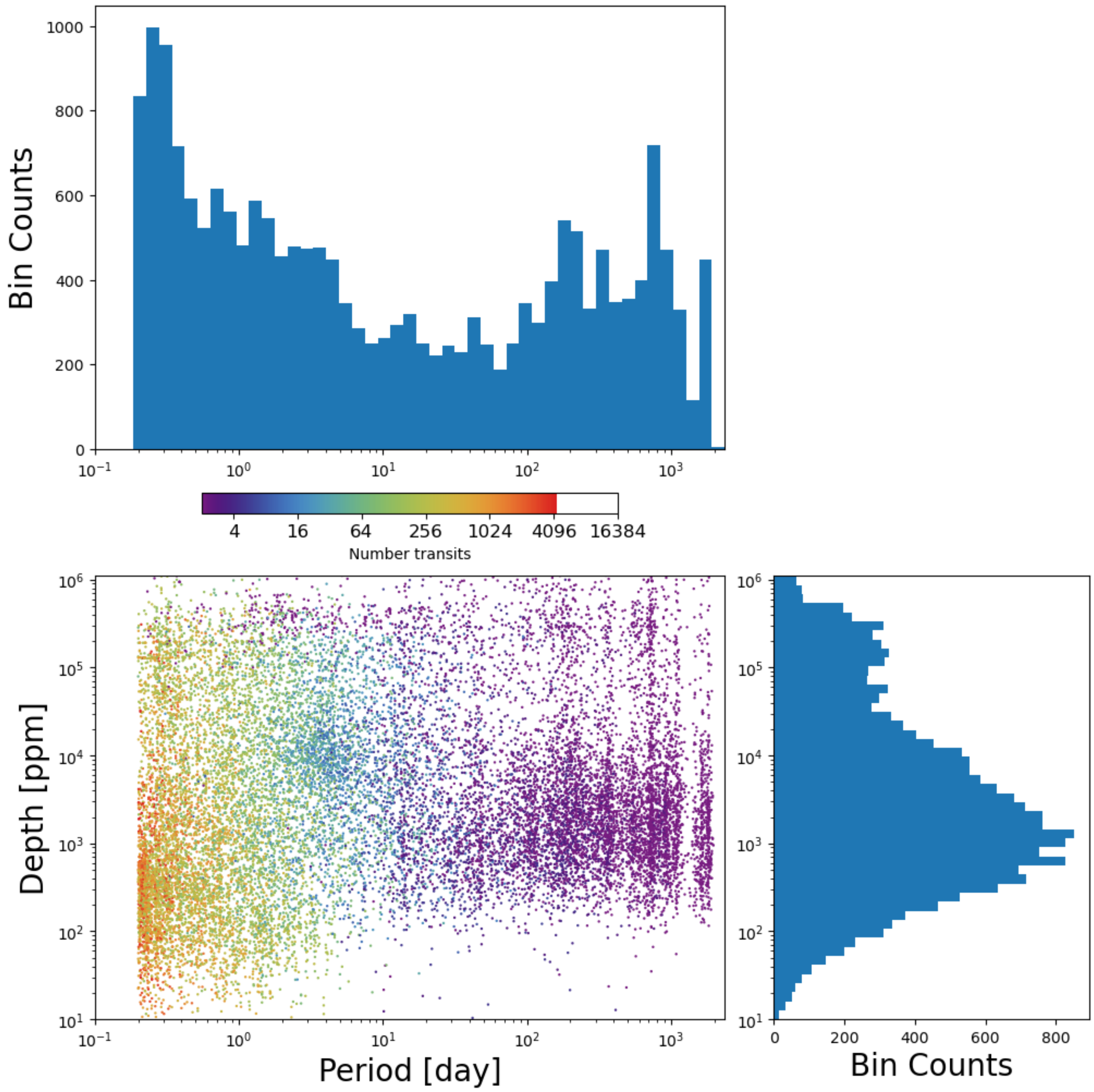

Figure 4: Lower Left Panel: Transit depth as a function of orbital period for the 19521 TCEs identified for the Sectors 14 – 86 multi-sector search. The color of the points indicates the number of transit events that were observed for the TCE. Reported depth comes from the DV limb darkened transit fit depth when available (or the DV trapezoid model fit depth if the limb darkened transit fit is not available). Top Panel: Orbital period distribution of the TCEs shown in the lower left panel. Right Panel: Transit depth distribution for the TCEs shown in the lower left panel.

Table 2: Targets in this Data Release With Number of Sectors Observed

Number of Sectors	Target Count
2	9867
3	6082
4	4666
5	2653
6	2371
7	1726
8	1473
9	1157
10	806
11	690
12	402
13	320
14	296
15	458
16	1000
17	361
18	305
19	303
20	288
21	314
22	291
23	360
24	535
25	739
26	751
27	385
28	183
29	196
30	195
31	156
32	100
33	66
34	56
35	59
36	89
37	101
38	155
39	202
40	201
41	159
42	119
43	45

Table 3: Sector 14 – 86 TCE Numbers

Number of TCEs	Number of Targets	Total TCEs
1	5601	5601
2	2568	5136
3	1088	3264
4	538	2152
5	274	1370
6	333	1998
–	10402	19521

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Twicken, J. D., Catanzarite, J. H., Clarke, B. D., et al. 2018, *PASP*, 130, 064502, doi: 10.1088/1538-3873/aab694

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Acronyms and Abbreviation List

BTJD Barycentric-corrected TESS Julian Date

CDPP Combined Differential Photometric Precision

Dec Declination

DV Data Validation Pipeline Module

KDPH Kepler Data Processing Handbook

MAST Mikulski Archive for Space Telescopes

MES Multiple Event Statistic

NAN Numerical Not-A-Number

POC Payload Operations Center

ppm Parts-per-million

RA Right Ascension

RMS Root Mean Square

SDPDD Science Data Products Description Document

SNR Signal-to-Noise Ratio

SPOC Science Processing Operations Center

TCE Threshold Crossing Event

TESS Transiting Exoplanet Survey Satellite

TIC TESS Input Catalog

TIH TESS Instrument Handbook

TJD TESS Julian Date

TOI TESS Object of Interest

TPS Transiting Planet Search Pipeline Module

UTC Coordinated Universal Time

XML Extensible Markup Language